

Complex oscillatory system

Y13 (2013) — English translation

The mass m_1 is connected by threads to two identical masses m_2 . The threads are thrown through small light blocks, the distance between which is L .

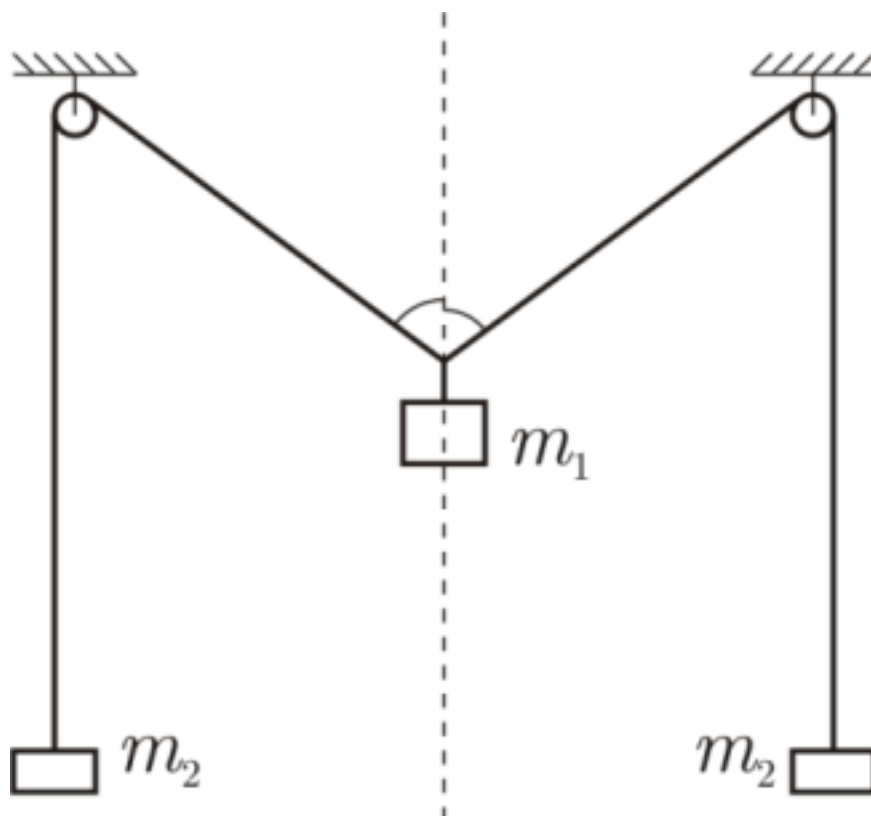


Figure 1: Figure 1.

A1	At what mass ratio m_1/m_2 will the system be in a stable equilibrium position (mass loads m_2 do not touch the blocks)?	0
A2	Find the period of small oscillations of such a system (the mass m_1 oscillates along a vertical straight line).	0
A3	Consider the special case of oscillations when $m_1 = m_2$.	0

Source: <https://pho.rs/p/3995>